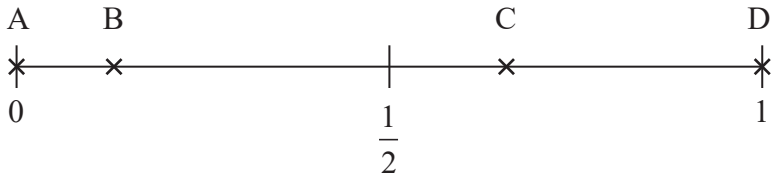
7 Here is a probability scale.
It shows the probability of each of the events A, B, C and D.



(a) Write down the letter of the event that is certain.

Certain has probability 1.

D

1 mark
D is at 1 on the scale

(1)

(b) Write down the letter of the event that is unlikely.

Unlikely: probability between 0 and one half.

B

1 mark
B is in this interval

(1)

There are 12 counters in a bag.

3 of the counters are red.
1 of the counters is blue.
2 of the counters are yellow.
The rest of the counters are green.

Caitlin takes at random a counter from the bag.

(c) Show that the probability that this counter is yellow or green is $\frac{2}{3}$

Green counters = 12 − 3 − 1 − 2 = 6

1 mark
Find the number of green counters

Yellow or green = 2 + 6 = 8 counters.

1 mark
Form 8/12: favourable outcomes divided by all 12 counters

Probability = $\frac{8}{12} = \frac{2}{3}$

1 mark
Simplify to show the stated result

(3)

(Total for Question 7 is 5 marks)